

Cardamo: An AI-Based Smart Cardamom Cultivation Management System Using Computer Vision and Predictive Analytics

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Abstract—To address these challenges, this study introduces Cardamo, an ML-driven smart cardamom cultivation management system designed to support Sri Lankan cardamom farmers through an integrated digital platform. The system incorporates four primary features which include cardamom leaf disease detection, smart pod monitoring, cardamom pod quality grading, and market price prediction with profit optimization. For the leaf disease detection and pod monitoring features, the system applies image pre-processing followed by EfficientNetB0-based classification, while the pod quality grading feature utilizes a specialized Keras-based CNN model to ensure objective assessment. Furthermore, the market prediction feature employs Ridge Regression to analyze complex variables such as historical prices, exchange rates, drying costs, and fresh to dried conversion factors to provide actionable profit-based recommendations. This integrated system addresses the current lack of a complete Artificial Intelligence (AI) based decision support platform for cardamom cultivation in Sri Lanka by combining advanced image recognition with predictive market analysis. Consequently, Cardamo supports early disease identification, pod condition monitoring, consistent quality grading, and strategic selling decisions through a unified interface. Through this comprehensive approach, the proposed system aims to reduce crop losses, improve export quality production, minimize manual errors, and support more profitable and sustainable cardamom cultivation.

Keywords—smart agriculture, cardamom cultivation, machine learning, computer vision, deep learning, explainable AI, quality grading, market price prediction

I. INTRODUCTION

Over the past years, the application of Computer Vision (CV), Machine Learning (ML), Deep Learning (DL), and predictive analytics has transformed the agricultural sector by introducing intelligent tools for crop monitoring, disease identification, quality assessment, and market-based decision-making. These technologies have improved traditional farming practices by enabling automated analysis of crop images,

early detection of plant health issues, and data-driven recommendations for farmers. In smart agriculture, such systems are especially important for high-value crops where product quality, disease control, harvesting decisions, and market timing directly affect farmer income and export performance. Cardamom, commonly known as the “Queen of Spices,” is one of the most valuable spice crops cultivated in Sri Lanka and plays an important role in the national spice export economy [1], [2]. It is mainly grown in humid highland regions such as Kandy, Matale, and Badulla, where environmental conditions strongly influence crop health, pod development, and final product quality.

However, cardamom cultivation faces several practical challenges, including leaf diseases, pod-related disorders, inaccurate maturity identification, inconsistent quality grading, and market price fluctuations. Previous studies on cardamom cultivation and agroecosystems have shown that pests, diseases, environmental conditions, and post-harvest handling can significantly affect crop yield, quality, and long-term sustainability [3], [4]. One of the main challenges faced by cardamom farmers is the early identification of leaf diseases. Common diseases such as leaf blight, leaf spot, mosaic disease can spread rapidly in plantation environments and reduce crop productivity. Traditionally, farmers identify these diseases through manual observation, which can be time-consuming, subjective, and inaccurate, especially when symptoms appear similar across different diseases.

DL-based image classification has shown strong performance in plant disease detection, where convolutional neural networks (CNNs) are used to identify disease patterns from leaf images [5], [6]. Foundational studies in DL and CNN based image recognition have further demonstrated the effectiveness of automated visual feature extraction for classifi-

cation tasks [7], [8]. However, many existing systems focus on general crops or laboratory image datasets, while limited research has specifically addressed cardamom leaf disease detection under real Sri Lankan plantation conditions. To address this requirement, the proposed system uses EfficientNetB0 as the image classification model for cardamom leaf disease detection. EfficientNet introduces a compound scaling method that balances network depth, width, and input resolution to improve both accuracy and computational efficiency [9]. Because EfficientNetB0 is the baseline and lightweight version of the EfficientNet family, it is suitable for image-based agricultural applications that require efficient deployment in web or mobile supported environments.

In the proposed system, cardamom leaf images are pre-processed using resizing, normalization, and noise reduction before being classified into disease categories. This enables the system to support early disease detection and assist farmers in taking timely crop management actions. In addition to leaf diseases, pod monitoring is important because the cardamom pod is the main commercial product of the plant. Pod related issues such as capsule borer damage, thrips damage, capsule rot, pod splitting and incorrect maturity identification can reduce both yield and market value. Manual pod inspection is difficult due to changes in lighting, background noise, pod color variation, and the similarity between some healthy and unhealthy pod appearances. Therefore, image based DL can support more consistent identification of pod conditions. In Cardamo, EfficientNetB0 is also applied for pod monitoring by learning visual patterns such as pod color, shape, surface texture, maturity appearance, and visible damage indicators. This allows the system to classify pod conditions and support better harvesting and caring decisions.

Cardamom pod quality grading is another major challenge in the industry. At present, grading is commonly performed manually based on visible characteristics such as size, color, shape, surface quality, and defects. This manual process can be subjective and may vary depending on the experience of the grader. Previous research has explored CV and ML techniques for cardamom grading, capsule size estimation, and sorting automation [10]-[14]. Other agricultural quality assessment studies have also shown that CNN based image classification can successfully identify quality related visual patterns in agricultural products [15], [16]. In the proposed system, a Keras based CNN model is used for pod quality grading to classify harvested cardamom pods into suitable quality grades based on visual features. Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM can be used to visually highlight the regions that influenced the model decision, improving transparency and user trust [17].

Apart from crop health and quality, cardamom farmers also face market related challenges. A major postharvest decision is whether to sell fresh cardamom pods immediately or dry them and sell later for a potentially higher price. This decision is influenced by historical price trends, seasonal production patterns, exchange rate fluctuations, drying costs, labor costs, and the fresh to dried weight conversion ratio. Previous studies

have shown that agricultural export prices can be affected by currency fluctuations and value addition costs in the spice sector [18], [19]. Time series forecasting and ML-based prediction methods have also been applied in agriculture to support market planning and decision making [20]-[22]. However, many forecasting approaches do not provide localized decision support for Sri Lankan cardamom farmers by comparing fresh selling and dried selling profitability.

For the market analysis component, the proposed system uses Ridge Regression to support price prediction and profit optimization. Ridge Regression is useful when multiple input variables are involved because it applies regularization to reduce overfitting and improve prediction stability in regression tasks [23]. In Cardamo, historical cardamom prices, exchange rate variations, seasonal trends, drying costs, labor costs, and processing expenses are considered as input features[24]. The model supports the prediction of market price behavior, while the profit optimization engine compares the expected income from selling fresh pods with the expected income from drying and selling later. Based on this comparison, the system generates a practical “Hold or Sell” recommendation for farmers.

To address these challenges, this study introduces Cardamo, an ML-driven smart cardamom cultivation management system designed to support Sri Lankan cardamom farmers through an integrated digital platform. The system incorporates four primary features: cardamom leaf disease detection, smart pod monitoring, cardamom pod quality grading, and market price prediction with profit optimization. The leaf disease detection and pod monitoring features utilize image preprocessing and EfficientNetB0-based classification, while the pod quality grading feature employs a Keras based CNN model. Additionally, the market prediction feature applies Ridge Regression to analyze historical prices, exchange rates, drying costs, and fresh to dried conversion factors to provide profit based recommendations. This integrated platform addresses the current lack of a complete Artificial Intelligence (AI) based decision support system for cardamom cultivation in Sri Lanka. By combining image classification based monitoring with predictive market analysis, Cardamo supports early disease identification, pod condition monitoring, consistent quality grading, and data driven selling decisions. Through these integrated features, the proposed system aims to reduce crop losses, improve export quality production, minimize manual errors, and support more profitable and sustainable cardamom cultivation.

II. LITERATURE REVIEW

In recent years, the integration of CV, DL, and predictive analytics has become increasingly important in smart agriculture for improving crop monitoring, disease detection, quality grading, and market-based decision making. A smart cardamom cultivation management system such as Cardamo requires the combined use of image classification, preprocessing techniques, interpretable DL, and predictive modeling. This literature review positions each component of the

proposed system within the current technological context by reviewing related work in cardamom cultivation, plant disease detection, pod monitoring, quality grading, and agricultural market prediction. Cardamom is one of the most valuable spice crops in Sri Lanka and contributes significantly to the national spice export sector [1], [2]. It is mainly cultivated in highland regions such as Kandy, Matale, and Badulla, where environmental conditions strongly influence crop productivity and quality. Studies on cardamom cultivation and agroecosystems highlight that pests, diseases, soil conditions, and post-harvest handling practices directly affect yield, sustainability, and export value [3], [4]. These findings show the need for intelligent agricultural systems that can support farmers through automated monitoring and decision-making. Leaf disease detection is a major area in which DL has shown strong potential. Traditional disease identification is usually performed through manual inspection by farmers or agricultural officers, which can be time-consuming and subjective. Previous studies have demonstrated that image-based plant disease detection using CNNs can identify disease symptoms such as discoloration, spots, and texture changes from leaf images [5], [6]. Foundational DL studies have further shown the effectiveness of CNN-based architectures in extracting complex visual features from image data [7], [8]. In addition, few-shot learning techniques have been explored to improve model performance when only a limited number of labelled images are available [25]. These studies provide the foundation for applying DL to cardamom leaf disease detection. For the leaf disease detection feature of Cardamo, EfficientNetB0 is selected as the main image classification model. EfficientNet-based architectures are designed to achieve better accuracy and computational efficiency by scaling network depth, width, and resolution in a balanced manner [9]. This makes EfficientNetB0 suitable for agricultural image classification tasks, especially when the system needs to perform efficiently in a web or mobile-based environment. In the proposed system, cardamom leaf images are first preprocessed through resizing, normalization, and noise reduction, and then passed into the EfficientNetB0 model to classify common cardamom leaf conditions such as healthy leaves, leaf blight, leaf spot, mosaic disease, chlorotic streak, and vein clearing. Pod monitoring is another important component because the cardamom pod is the main commercial product of the plant. Pod-related issues such as capsule borer damage, thrips damage, capsule rot, pod splitting, shrivelling, and incorrect maturity identification can reduce both yield and market value. Manual pod monitoring is difficult because visual symptoms may vary depending on pod maturity, lighting conditions, background noise, and field environment. Instead of relying only on manual inspection, DL-based image classification can support more consistent pod condition identification [26], [27]. In the proposed system, EfficientNetB0 is used for pod monitoring because it can extract important visual patterns such as pod color, surface texture, shape variation, and visible defect patterns. By applying this model to pod images, the system can classify pod conditions and support farmers in identifying maturity or damage-related issues. Automated

cardamom pod quality grading has also been studied using CV and ML techniques. Traditional grading depends on visible features such as pod size, color, shape, texture, and surface defects. However, this process is subjective and may vary between graders. Previous studies on cardamom grading have shown that CV can be used to classify pods into different quality categories [11], while other research has combined CV with additional sensing methods to improve grading accuracy [12]. Image processing approaches have also been used to estimate cardamom capsule size, surface area, and physical characteristics, showing the importance of geometric features in grading [13]. ML-based grading and sorting systems have further demonstrated the possibility of automating cardamom quality assessment [10], [14]. These studies support the development of a DL-based grading component in Cardamo. For the pod quality grading component, Cardamo uses a Keras-based CNN model to classify harvested cardamom pods into quality grades based on visual appearance. The model analyzes features such as color uniformity, pod shape, surface quality, and visible defects. Related agricultural studies have shown that CNN-based image classification and defect detection approaches can successfully identify quality-related visual patterns in agricultural products such as fruits and spices [15], [16]. By using EfficientNetB0, the grading component aims to provide a lightweight and accurate classification approach that can support farmers, traders, and exporters in making more consistent quality assessments. To improve transparency, XAI techniques such as Grad-CAM can be integrated to highlight image regions that influenced the model's grading decision [17]. Market prediction and profit optimization are critical for farmers facing uncertainty in selling prices. Decisions depend on historical price trends, currency exchange rates, and drying costs [18], [19]. Currency fluctuations significantly influence local market prices [20]. ML-based prediction methods support market planning [21]-[29]. Ridge Regression is suitable for such tasks because it handles multiple input variables while reducing overfitting [23]. The model supports decision-making by considering historical prices and seasonal trends [24]. Furthermore, spice crop markets benefit from machine learning techniques [30], and decision support systems improve post-harvest management [31].

For the market analysis component, Ridge Regression is used as the predictive model. Ridge Regression is suitable for market prediction tasks because it can handle multiple input variables while reducing overfitting through regularization [23]. In the proposed system, factors such as historical cardamom prices, exchange rate changes, seasonal trends, drying costs, and processing expenses are used as input features. The model predicts possible market price behavior and supports the profit optimization engine. The system then compares the expected profit from selling fresh cardamom immediately with the expected profit from drying and selling later. By considering the fresh-to-dried weight conversion ratio and processing costs, the system generates a practical "Hold or Sell" recommendation for farmers. Based on the reviewed literature, most existing systems focus on individual

agricultural tasks such as plant disease detection, crop grading, or price prediction. However, there is limited research on an integrated cardamom-specific platform that combines leaf disease detection, pod monitoring, pod quality grading, and market profit analysis. Therefore, there is a clear need for a unified system that uses efficient DL and predictive modelling techniques to support Sri Lankan cardamom farmers. The proposed Cardamo system addresses this gap by integrating EfficientNetB0-based image classification for leaf disease detection and pod monitoring, Keras-based CNN classification for pod quality grading, with Ridge Regression-based market analysis and profit optimization. This combination provides a practical, user-friendly, and data-driven smart agriculture solution. By supporting early disease identification, pod condition monitoring, consistent quality grading, and “Hold or Sell” market recommendations, Cardamo contributes to reducing crop losses, improving export-quality production, and increasing farmer profitability in the Sri Lankan cardamom industry.

III. METHODOLOGY

The implementation of Cardamo follows a structured methodology that combines image processing, DL, ML, and predictive analytics to support smart cardamom cultivation management. The methodology explains the steps and techniques used to develop Cardamo, an ML-driven system for cardamom leaf disease detection, pod monitoring, pod quality grading, and market-based decision support. The major stages of the methodology include data collection, data preprocessing, model development, system development, evaluation and validation, and technology stack selection.

A. Data Collection and Preprocessing

Data collection was specifically tailored to the functional requirements of the system’s four primary features. The visual dataset consisted of cardamom leaf images for disease detection, pod images for monitoring maturity and disorders, and harvested cardamom pod images for automated quality grading. For the market analysis feature, numerical data including historical cardamom prices, exchange rates, and processing costs were sourced from established records to support profit comparisons and “Hold or Sell” recommendations [22], [24].

For the image-based features, all raw data underwent a multi stage pre-processing pipeline. The images were resized to a standardized resolution of 224×224 pixels and normalized to ensure uniform feature extraction. To improve model generalization in realistic field conditions, data augmentation techniques including 20 degree rotations, horizontal/vertical flips, and brightness adjustments were applied to the training set. For market data, a cleaning process was implemented to address missing values and normalize numerical features for regression modeling.

B. Model Development

The core of the Cardamo system consists of four specialized models, each developed to address a specific challenge in

cardamom cultivation. We used different machine learning and deep learning techniques based on the nature of each feature.

- 1) *Leaf Disease Detection*: Utilized the EfficientNetB0 model to identify common cardamom leaf pathologies like leaf blight and mosaic disease. We selected this model because it is very accurate but lightweight, which makes it suitable for mobile-based agricultural applications [9].
- 2) *Smart Pod Monitoring*: This feature also employs EfficientNetB0 to classify pod images based on maturity and health conditions. The model identifies maturity stages and detects disorders like capsule borer damage and cardamom thrips, assisting farmers in harvesting decisions.
- 3) *Quality Grading*: We developed a custom CNN using Keras to automate the grading of harvested pods. The model analyzes visual features such as color uniformity, shape, and surface texture to classify pods into consistent quality grades, reducing manual subjectivity.
- 4) *Market Prediction*: For market analysis, we applied Ridge Regression to predict market price behavior and support profit-based decision-making [23]. The model evaluates several input factors, including historical prices and exchange rate variations. By comparing the potential profit from selling fresh versus dried pods, the system provides “Sell Fresh” or “Hold and Dry” recommendations.

C. System Integration and Workflow

The four specialized models are integrated into a unified digital platform to provide a comprehensive decision-support tool for cardamom farmers. As illustrated in the system architecture (Fig. 1), the workflow is initiated when a user submits an image or enters numerical data through the interface. For image-based features, the backend automatically performs the standardized preprocessing pipeline before routing the data to the corresponding EfficientNetB0 or CNN model. In the case of market analytics, the input parameters are processed through the Ridge Regression model to evaluate profitability. All analytical outputs, processed images, and historical records are maintained in a centralized database, allowing for continuous monitoring of cultivation progress. To enhance transparency, Explainable AI (XAI) via the Grad-CAM technique is utilized within the workflow. When a classification or grading decision is made, the system generates a heatmap overlaid on the original image. This visually identifies the specific regions, such as necrotic spots on a leaf or surface defects on a pod, that influenced the model’s output. By providing these visual justifications, the system ensures a transparent decision-making process, which is essential for building user trust in AI-driven agricultural tools.

D. Evaluation and Validation

The evaluation of Cardamo was conducted independently for each system feature to ensure technical accuracy and practical reliability. For the image-based features, the datasets

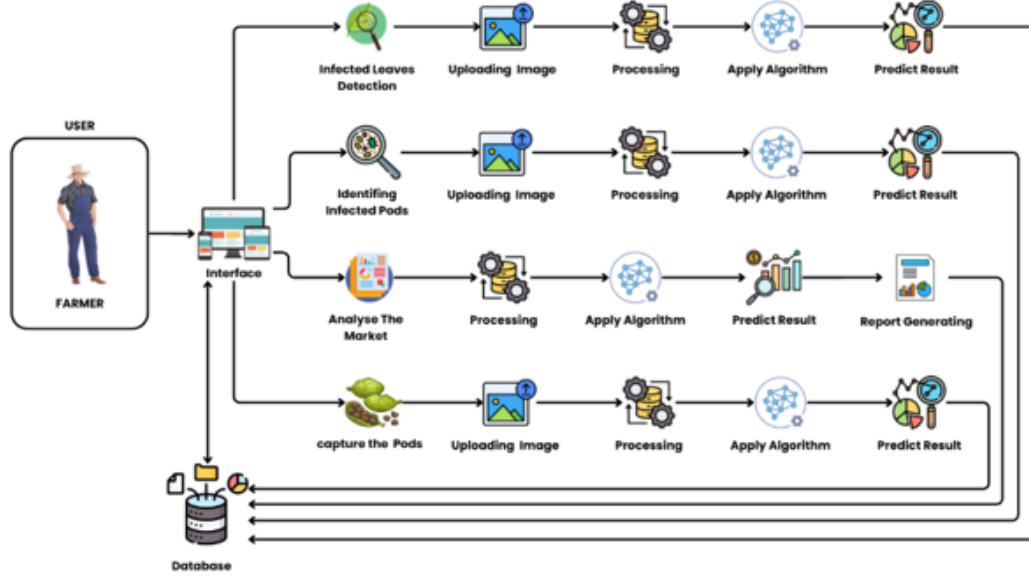


Fig. 1. System architecture diagram

were partitioned into training, validation, and testing sets using an 80:10:10 ratio. The training and validation sets were utilized for model optimization, while the testing set provided an objective assessment of performance on unseen data. The leaf disease detection, pod monitoring, and quality grading features were evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-score. Confusion matrices were generated to identify specific misclassification patterns between disease types, pod conditions, and maturity stages. The market analysis feature was assessed using regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R square score. These metrics measured the deviation between predicted and actual market values. Furthermore, the "Hold or Sell" recommendation logic was validated by cross-referencing system outputs with historical market trends and expected profit outcomes. Qualitative validation was also performed by comparing predicted labels with visible field conditions and testing sample farmer inputs to ensure the practical viability of the decision-support results.

IV. RESULTS AND DISCUSSION

The Cardamo system was implemented using Python, with TensorFlow and Keras for DL components, and Scikit-learn for predictive analytics. The system was evaluated using standard classification and regression metrics to ensure the reliability of each component.

A. Leaf Disease Detection Results

The EfficientNetB0 model for leaf disease detection achieved a classification accuracy of **87%**. The model successfully identified symptoms of leaf blight, mosaic, and chlorotic streak. Preprocessing techniques like noise reduction significantly enhanced the model's ability to distinguish between similar disease patterns in complex plantation backgrounds

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Feature	Model	Accuracy	F1-Score
Leaf Disease	VGG16	79%	0.78
	EfficientNetB0 (Ours)	87%	0.86
Pod Monitoring	ResNet50	81%	0.80
	EfficientNetB0 (Ours)	87%	0.85
Quality Grading	Random Forest	75%	0.74
	Keras-based CNN (Ours)	86%	0.84
Market Analysis	Linear Regression	72%	-
	Ridge Regression (Ours)	81%	-

B. Smart Pod Monitoring Results

The pod monitoring component demonstrated high precision in identifying maturity stages and disorders. It achieved an accuracy of **87%** in detecting capsule borer damage, thrips damage, and capsule rot. XAI through Grad-CAM confirmed that the model effectively focused on localized pod defects and surface textures for its predictions.

C. Quality Grading Results

The Keras-based CNN model for quality grading achieved an accuracy of **86%**. By analyzing size, color uniformity, and surface defects, the model provided objective grading that outperformed traditional manual inspection. This automation reduces subjectivity and ensures consistent quality standards for the export market.

D. Market Prediction and Profit Optimization

The Ridge Regression model was evaluated using MAE and R^2 values. The model provided stable price forecasts despite market volatility. The profit optimization engine was validated against historical data, confirming that the "Hold or Sell" recommendations accurately identified the most profitable selling

windows for farmers by considering fresh-to-dried conversion ratios and processing costs.

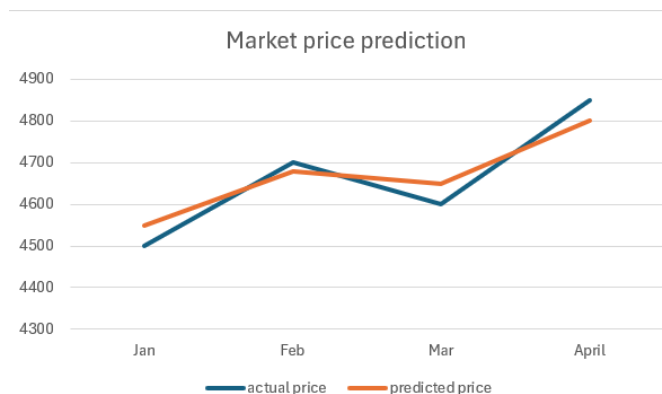


Fig. 2. Actual vs. Predicted market prices using Ridge Regression

V. CONCLUSION

The Cardamo system offers a comprehensive decision-support platform for cardamom cultivation by integrating image processing, DL, and predictive analytics. It addresses critical challenges including leaf diseases, pod monitoring, manual grading limitations, and price uncertainty. Utilizing EfficientNetB0 and Keras-based CNNs, the system provides accurate disease detection and automated quality grading. Furthermore, Ridge Regression optimizes market decisions via “Hold or Sell” recommendations based on fresh and dried cardamom profitability. By enabling early disease detection and data-driven insights, Cardamo helps reduce crop losses and improve export quality. Future work will focus on expanding datasets, enhancing model accuracy, adding XAI, and field-validating the platform across Sri Lankan cardamom-growing regions.

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